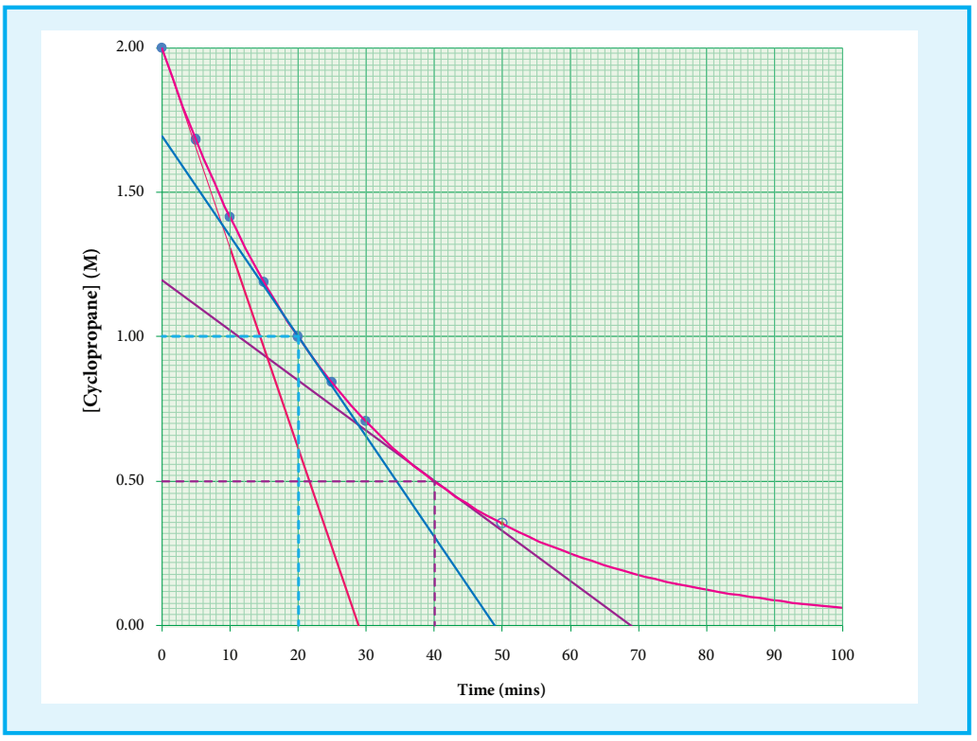


Fig 7.2 Concentration of cyclopropane vs time - graph

Let us calculate the instantaneous rate of isomerisation cyclopropane at different concentrations: 2 M, 1M and 0.5 M from the graph shown in fig 7.2, the results obtained are tabulated below.

[cyclopropane] mol L^{-1}	Rate mol $L^{-1}min^{-1}$
2	6.92×10^{-2}
1	3.46×10^{-2}
0.5	1.73×10^{-2}

Table 7.2 Rate of isomerisation

7.3 Rate law and rate constant:

We have just learnt that, the rate of the reaction depends upon the concentration of the reactant. Now let us understand how the reaction rate is related to concentration by considering the following general reaction.

$$xA + yB \longrightarrow \text{products}$$

The rate law for the above reaction is generally expressed as

$$\text{Rate} = k\,[A]^m[B]^n$$

Where k is proportionality constant called the rate constant. The values of m and n represent the reaction order with respect to A and B respectively. The overall order of the reaction is given by (m+n). The values of the exponents (m and n) in the rate law must be determined by experiment. They cannot be deduced from the Stoichiometry of the reaction. For example, consider the isomerisation of cyclopropane, that we discussed earlier.

The results shown in table 7.2 indicate that if the concentration of cyclopropane is reduced to half, the rate also reduced to half. It means that the rate depends upon [cyclopropane] raised to the first power

i.e., $\text{Rate} = k[\text{cyclopropane}]^1$

$$\Rightarrow \frac{\text{Rate}}{[\text{cyclopropane}]} = k$$

XII U7 kinetics - Jerald Folder.indd 208 04-01-2021 13:31:49